Department of ICT
Faculty of Technology
University of Ruhuna

Data Structures and Algorithms – ICT2113 **Level 2- Semester 1**

Laboratory Assignment 8

Objectives:

Understand the implementation of sorting techniques using C.

1. I) Write a function **bubble_sort(arr[], num)** which use to sort the following sample data using bubble sort algorithm.

Sample data : 100, 22, 55, 12, 75, 80, 150, 10, 99, 40

```
int main() {
  int arr[10], num, i;

  printf("\nEnter no of elements :");
  scanf("%d", &num);

  printf("\nEnter array elements :");
  for (i = 0; i < num; i++)
    scanf("%d", &arr[i]);

  bubble_sort(arr, num);
}
}
```

II) Calculate the average marks using separate function called **float avgmarks(arr[], num)** and display the above sample data which are greater than the average mark.

2. Write a function **selectionsort(arr,num)** which sort the array **arr** with **num** elements. Use the **selection sort** algorithm for this sorting.
    a. Read the elements from the keyboard.
    b. Sort these elements.
    c. Display the sorted list.
    d. Modify above function to sort data in descending order

3. Write a function **insertsort(arr, num)** which sort the array **arr** with **num** elements. Use the insertion sort algorithm for this sorting.
    a. Read the elements from the keyboard.
    b. Sort these elements.
    c. Display the sorted list.
    d. Modify above function to sort data in descending order

4. Write a C program to sort a string in alphabetic order

    For example, if a user will enter a string "programming" then the output will be "aggimmnoprr" or output string will contain characters in alphabetical order.

    Assume input string contains only lower case alphabets.

5. The following words need to be sorted,

    information engineering biosystem mobile technology

    a. In alphabetical (a to z) order.
    b. In the reverse alphabetical (z to a) order.

    These words should store in an array. Write methods to implement sorting of these char arrays using Bubble Sort algorithm.